WORKED SOLUTION

STEP III 2010, Question 4

Pure • Common Roots of Polynomials

STEP III 2010, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q4

The key idea

If α satisfies two quadratics with the same leading coefficient, it also satisfies their difference, and the x^2 terms cancel to leave a *linear* equation. That pins α down, and putting it back into one quadratic gives the condition. In part (ii) the same trick works once we divide the cubic by the quadratic: on the roots of the quadratic, the cubic behaves exactly like its remainder, so we are back to two quadratics and can quote part (i).

Solution

Throughout, "root" means a root in the complex numbers, so every quadratic has two roots (counted with multiplicity).

Part (i)

Suppose α satisfies both equations. Subtracting the second from the first,

$$(a-c)\alpha + (b-d) = 0.$$

Since $a \neq c$ we may divide by $a-c$, giving

$$\alpha = -\frac{b-d}{a-c},$$

as required.

The condition. Write $k = a - c \neq 0$ and $m = b - d$, so the condition is $m^2 - amk + bk^2 = 0$.

($\Rightarrow$) If there is a common root, it is $\alpha = -m/k$ by the first result. Substituting into $\alpha^2 + a\alpha + b = 0$,

$$\frac{m^2}{k^2} - \frac{am}{k} + b = 0,$$

and multiplying by $k^2 \neq 0$ gives $m^2 - amk + bk^2 = 0$.

($\Leftarrow$) Suppose $m^2 - amk + bk^2 = 0$, and *define* $\alpha = -m/k$. Dividing the condition by k^2 and reading the step above backwards shows that $\alpha^2 + a\alpha + b = 0$. Also $(a-c)\alpha + (b-d) = k\alpha + m = 0$ by the choice of α. So

$$\alpha^2 + c\alpha + d = (\alpha^2 + a\alpha + b) - ((a-c)\alpha + (b-d)) = 0 - 0 = 0,$$

and α is a common root. This proves the equivalence when $a \neq c$.

? **Ask yourself** Why isn't the forward direction enough on its own?

The forward direction only says that *if* a common root exists, the condition holds. For "if and only if" we must also show that the condition produces a common root. That is why we defined α from the condition and checked that it satisfies *both* equations.

Dropping $a \neq c$. Suppose $a = c$. Then the condition reads $(b-d)^2 = 0$, that is, $b = d$.

- If $b = d$, the two equations are identical, so they certainly have a common root (both of their roots are common), and the condition holds.
- If $b \neq d$, a common root α would give $(a-c)\alpha + (b-d) = b - d = 0$ on subtracting, which is a contradiction. So there is no common root, and the condition fails.

In both cases "common root" and "condition" agree, so **yes, the result still holds** without the condition $a \neq c$.

(The one subtlety is when only real roots count. If $a = c$, $b = d$ and $a^2 < 4b$, the condition holds, yet the identical equations have no real roots. When $a \neq c$ there is no such problem, since for real coefficients the common root $-(b-d)/(a-c)$ is automatically real.)

Part (ii)

Divide the cubic by $x^2 + ax + b$. Since $(x+1)(x^2 + ax + b) = x^3 + (a+1)x^2 + (a+b)x + b$,

$$x^3 + (a+1)x^2 + qx + r = (x+1)(x^2 + ax + b) + (q - a - b)x + (r - b).$$

So if $x^2 + ax + b = 0$ at $x = \alpha$, the cubic takes the value $(q - a - b)\alpha + (r - b)$ there, and so does the quadratic

$$(x^2 + ax + b) + (q - a - b)x + (r - b) = x^2 + (q - b)x + r.$$

Hence α is a common root of the quadratic and the cubic if and only if it is a common root of

$$x^2 + ax + b = 0 \qquad \text{and} \qquad x^2 + (q-b)x + r = 0.$$

Now apply part (i) with $c = q - b$ and $d = r$. Then $a - c = a + b - q$ and $b - d = b - r$, so there is a common root if and only if

$$(b-r)^2 - a(b-r)(a+b-q) + b(a+b-q)^2 = 0.$$

Because the part (i) result holds whether or not $a = c$, no extra condition is needed here.

The example. Divide both equations by 2 to make them monic:

$$x^2 + \tfrac{5}{2}x + b = 0, \qquad x^3 + \tfrac{7}{2}x^2 + \tfrac{5}{2}x + \tfrac{1}{2} = 0.$$

These have the required form with $a = \frac{5}{2}$ (and indeed $a + 1 = \frac{7}{2}$), $q = \frac{5}{2}$ and $r = \frac{1}{2}$. Then $a + b - q = b$ and $b - r = b - \frac{1}{2}$, so the condition becomes

$$\left(b - \tfrac{1}{2}\right)^2 - \tfrac{5}{2}\left(b - \tfrac{1}{2}\right)b + b \cdot b^2 = 0.$$

Multiplying by 4:

$$\begin{aligned}(2b-1)^2 - 5b(2b-1) + 4b^3 &= 0\\ 4b^3 - 6b^2 + b + 1 &= 0.\end{aligned}$$

By inspection $b = 1$ is a root, and

$$4b^3 - 6b^2 + b + 1 = (b-1)(4b^2 - 2b - 1).$$

The quadratic factor gives $b = \dfrac{2 \pm \sqrt{4+16}}{8} = \dfrac{1 \pm \sqrt{5}}{4}$. So the equations have a common root exactly when

$$b = 1 \qquad \text{or} \qquad b = \frac{1+\sqrt{5}}{4} \qquad \text{or} \qquad b = \frac{1-\sqrt{5}}{4}.$$

Otherwise. The cubic here has no b in it, so we can just solve it. Trying $x = -\frac{1}{2}$ gives $-\frac{1}{4} + \frac{7}{4} - \frac{5}{2} + 1 = 0$, so

$$2x^3 + 7x^2 + 5x + 1 = (2x+1)(x^2 + 3x + 1),$$

with roots $-\frac{1}{2}$ and $\frac{1}{2}(-3 \pm \sqrt{5})$. A common root x forces $2b = -(2x^2 + 5x)$. For $x = -\frac{1}{2}$ this gives $b = 1$. For the other two roots, $x^2 = -3x - 1$, so $2x^2 + 5x = -x - 2$ and $b = \frac{1}{2}(x+2) = \frac{1}{4}(1 \pm \sqrt{5})$. This agrees with the answer above.

Checks

- With $b = 1$, the quadratic $2x^2 + 5x + 2 = (2x+1)(x+2)$ has the root $-\frac{1}{2}$, which is a root of the cubic.
- $b = \frac{1}{4}(1+\sqrt{5}) \approx 0.8090$: the quadratic's roots are ≈ -0.3820 and -2.1180, and $-0.3820 \approx \frac{1}{2}(-3+\sqrt{5})$ is a root of the cubic.
- A computer test of 20 000 random integer choices of a, b, c, d (and of a, b, q, r) found that the condition vanishes exactly when the equations share a complex root, including the cases with $a = c$.

Common slips

- Proving only "common root $\Rightarrow$ condition" and forgetting the converse.
- Answering the $a = c$ question with "no, we divided by $a - c$". The proof breaks down, but the *result* survives, because both sides reduce to $b = d$.
- Applying part (ii) without first dividing by 2: the formula needs monic equations.

Beyond STEP

The expression in part (i) is the *resultant* of the two quadratics: a polynomial in the coefficients that vanishes exactly when the polynomials share a root. Resultants come from Sylvester's determinant, and they underlie the elimination theory used to solve systems of polynomial equations. The "divide and keep the remainder" step in part (ii) is the first step of the Euclidean algorithm for polynomials, which finds the highest common factor of two polynomials just as it does for integers. Allenby's *Rings, Fields and Groups* develops polynomial division and highest common factors in this setting.

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